Department of Economics
Jahangirnagar University
M.Sc. Final Examination 2023
Course No: 501 (Advanced Microeconomic Theory)
Full Marks: 70 Time: Four Hours

*Answer any **five** of the following questions.*
All parts of a question must be answered consecutively

1. a) Assume there are only two goods, X and Y. Prove that quasi-concave utility $U(X, Y)$ is identical to $dMRS/dX < 0$ (3)

b) After grading exams, your instructor only obtains utility from his favorite cocktail- the Negroni. Each Negroni consists of 1 ounce of Gin, 1 ounce of red Vermouth, and 1 ounce of Campari. Let G stand for ounces of Gin, V stands for ounces of Vermouth, and C stands for ounces of Campari. Each good also has a price denoted by P_g, P_v, and P_c, respectively.

i) Write your instructor's utility function. (2)

ii) Find his Marshallian demand for Gin. (2)

iii) Find his Hicksian demand for Campari (2)

iv) Find the expenditure function (2)

c) $U = X + ln(Y)$. Find the demands for X and Y. Suppose you only have $1¢$. Which goods do you buy? Why? (3)

2. a) A person's preferences are presented by the utility function $U = X + log(Y)$ (as always, the log is the base e or natural log). Does this person have convex indifference curves? Provide proof. (2)

b) The following is the log of an *individual's* demand for good X as a function of prices and income

$$X(P_x, P_y, P_z, I) = \alpha P_x^{\alpha} P_y^{\beta} P_z^{\gamma} I^{\theta}$$

i) What is the cross-price elasticity with good Z? (2)

For parts ii, *iii*, and iv, use the following values for the parameters:

$$\alpha = -0.5 ; \gamma = 0.4 ; \theta = 0.8$$

ii) What is income elasticity? Is X a luxury, necessity, or an inferior good? (2)

iii) Is good Z a gross substitute or a gross complement for good X? (2)

iv) If the price of Y were to fall by 5%, by what percent and in what direction would the quantity demanded of X change? (2)

v) If the price of good X decreases, would this person's expenditure on X increase, decrease, or remain the same? (2)

c) An agent has preferences for goods X and Y represented by the utility function

$$U(X, Y) = X + 3Y$$

The price of good X is $P_x = 20$, the price of good Y is $P_y = 40$, and her income is $I = 400$. Choose the quantities of X and Y that maximize her utility for the given prices and income. (2)

3 a) You are hired by Burger King to estimate the demand for its Whopper sandwich. Initially, you choose a very simple demand function, and the following estimate

$$Q = 10 + 1.3P_w + 0.4P_{bw} - 3P_f + 2.2\,I$$

Where, Q is the quantity of Whoppers (in thousands), P_w is the price of a Whopper, P_{bw} is the price of a McDonald's Big Mac (McDonald's is Burger King's largest competitor), P_f is the price of an order of French fries at Burger King (all prices are in \$'s), and I is the average income of the region (in thousands of \$'s).

i) You notice immediately that there is a terrible problem with this estimate. What is the problem? (2)

b) To correct the problem, you change the structural form of your demand function and add one additional variable. The correctly estimated demand function is:

$$Q = 150P_w^{-1.5}P_{r1}^{0.75}P_{r2}^{-0.25}I^{0.8}A^{0.05}$$

Where, Q, P_w, and I are defined as before, P_{r1}, and P_{r2} are the prices of related goods (in \$'s), and A is advertising expenditure (in thousands of \$'s). Use this correct demand function to answer the following questions.

i) As before, the two related goods are McDonald's Big Macs and Burger King French fries. Which related good (i.e., $r1$ or $r2$) is French fries? (2)

ii) Is the Burger King Whooper a luxury, necessity, or inferior good? (2)

iii) Suppose Burger King *increases* its advertising expenditure by 50%. By what percentage and in what direction will the demand for Whoppers change? (2)

iv) Burger King decides to lower the price of its French fries by 50%. By what percentage and in what direction (i.e., increase or decrease) will the quantity of Whoppers sold change? (2)

v) Suppose the marginal cost of making a Whopper is \$1.00. What is the profit-maximizing price of a Whopper? (2)

vi) McDonald's runs a sale and lowers its price of Big Macs by 10%. By what percentage must Burger King lower its price to not lose any customers. (2)

4. a) Below are intuitive explanations of four properties of cost functions. Please write the formal mathematical property next to the correct intuitive explanation. You can use w_i and l_i to represent generic input prices and inputs, respectively.

i) If all prices *increase*, the cost of producing a given output level must increase. (1.5)

ii) If all prices *increase* by the same proportion, costs must increase by that same proportion. (1.5)

iii) If prices *increase* but relative prices change, the input mixture will change, and costs will not go up by as much as they would if the input mix remained the same. (1.5)

iv) If it costs more to make more. (1.5)

b) A sausage maker uses three ounces of pork and two ounces of veal to make one 5-ounce sausage. He pays 30 cents for an ounce of pork and 80 cents for an ounce of veal.

i) Let P stand for ounces of pork, V stands for ounces of veal, and S stands for a number of sausages. Write the production function for sausages. (2)

ii) Find the conditional factor demand for pork. (2)

iii) What is the cost function for sausages? (2)

iv) Suppose the demand for sausages is given by $Q = 200P^{-6}$. What is the profit-maximizing price to charge for sausages? (2)

5. a) A firm's technology is represented by the production function $Q = (KL)^{1/3}$

In the short run, K is fixed at $\overline{K} = 64 = 4^3$

i) What is the firm's short-run production function? (1)

ii) Find the short-run conditional factor demand for L. (1)

iii) What is the short-run cost function? (1)

iv) What is the shutdown price? (1)

v) Assume the firm is a price taker in output (let P be the output price). Find the firm's short-run supply. (1)

Assume now we are in the long run, and K is no longer fixed

vi) What are the returns to scale for this production function? (1)

vii) Calculate the elasticity of substitution σ. (1)

viii) Find the long-run conditional factor demands for L and K. (1)

ix) Find the long-run cost function. (1)

x) Again, assuming the firm is a price taker, find the long-run supply. (1)

b) Starbucks decides to introduce a new "Jumbo" iced tea, which will come in a 62-ounce cup. The current Trenta cup holds 31 ounces. Will the Jumbo Cup cost twice as much, more than twice as much, or less than twice as much as the Trenta Cup? Please explain why you chose your answer. (4)

6. a) A competitive industry consists of 1,000 identical firms, each with short-run total costs given by $C_{SR}(q) = 0.5q^2 + 2q + 80$.

i) What is the short-run supply for each firm? (1)

ii) What is the industry short-run supply? (1)

Suppose market demand is $Q = 44{,}000 - 1000P$

iii) Find the equilibrium price and quantity. (2)

iv) What is each firm's short-run profit? (2)

v) Suppose the government adds a \$20 per unit tax. Find the new equilibrium price demanders pay, the price suppliers keep, and the quantity produced. (2)

vi) Calculate the welfare cost of the tax. (2)

b) In a three-good world, the excess demand for good 2 is given by

$$Z_2(P) = 4 - \frac{P_2}{P_1} + \frac{2P_3}{P_1}$$

and the excess demand for good 3 is given by

$$Z_3(P) = 8 - \frac{P_2}{P_1} + \frac{4P_3}{P_1}$$

i) Find the excess demand function for good 1. (1)

ii) What are the market-clearing price ratios $\frac{P_2^*}{P_1^*}$, $\frac{P_3^*}{P_1^*}$ and $\frac{P_2^*}{P_3^*}$? (3)

7. The domestic demand for color printers is $Q = 250{,}000 - 100P$. The domestic supply is $Q = 250P - 30{,}000$

a) Suppose no international trade takes place. Find the domestic autarky price and quantity. (2)

b) Suppose this country can import or export as many printers as it wants at the world price of $P_w = 600$. Would the country import or export printers? How many would it import/export under free trade? (3)

c) Suppose the domestic government puts a tariff of $50 per imported printer. What would be the new market equilibrium (i.e., P; domestic q_d and q_s; and imports)? How much tariff revenue would the government gain? (3)

d) Assuming that supply = MC, what are the welfare effects of the tariff (i.e., ΔCS; ΔPS; ΔW)? (3)

e) Assume instead that the exporting nation signs a Voluntary Export Restraint with the domestic government that limits exports to 20,000 printers. Find the welfare effects of the VER as compared to free trade. (3)

8. a) Explain Hicks-Kaldor compensation principle of attaining social welfare. How did Scitovski correct the contradiction arises in this principle. (6)

b) Derive the grand utility possibility frontier and determine the 'point of bliss'. Does such point ensure maximum social welfare? Justify your explanation. (8)

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Course No: 502 (Advanced Macroeconomic Theory)
Full Marks: 70 Time: Four Hours

*Answer any **five** of the following questions.*
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1. Solow model focuses on four variables like output (Y), capital (K), labor (L) and knowledge or "effectiveness of labor" (A). The growth rate of X refers $\dot{X}/X$; the growth rate of L is n; growth rate of A is g and capital depreciation rate is δ. [4+3+3+4]
   a) Briefly discuss the assumptions about input, output and production function to construct Solow exogenous growth model. What is the production function of capital augmenting and Hicks-neutral technological progress?
   b) Briefly describe the dynamics of k (capital). Why does it is called the Balanced Growth Path (BGP)?
   c) Describe how each of the following affects the break-even and actual investment in the basic diagram for the Solow model:
      i) Increase in the rate of capital depreciation.
      ii) Improvement in the rate of technological progress.
      iii) Capital's share (α) rises for an intensive form of production function.
   d) Show the impacts of savings ratio in the steady state solution of Solow growth model and prove that exogenous changes in the savings ratio leave only level effects on k and y, but not on growth effects.

2. In the Romer model of endogenous technological progress, the process of technological progress is given by $\dot{A} = A0.5LR$, where LR is the share of researchers in the population. The population grows at the rate n. [4+4+3+3]

   a) Find the steady state level of technological progress. Suppose that LR will increase by 10% (increased by 1.1 times).
   b) Calculate the effect of this change on technological progress at the moment of shock (on impact).
   c) Calculate its effect on the steady state level of technological progress.
   d) Illustrate your answers on the main diagram of the model (g^A on the horizontal axis, $\dot{g}^A/g^A$ on the vertical axis.

3. Consider a Ramsey-Cass-Koopmans model with the production function $Y = F(K, AL)$ and real interest rate r; household's utility function takes the form [4+4+2+4]

$$U = \int_{t=0}^{\infty} e^{-\rho t} u(c(t)) \frac{L(t)}{H} dt$$

and the instantaneous utility function takes the CRRA (constant –relative –risk-aversion) utility form $U(c(t)) = \frac{C(t)^{1-\theta}}{1-\theta}$; Where θ determine the household's willingness to shift consumption between different periods.
   a) Define Euler equation. Find out the Euler equation for households' maximization problem and interpret it.
   b) Describe the phase diagram in terms of the evolution of c and k.
   c) What is the difference between the balanced growth paths of the Solow and Ramsey–Cass–Koopmans on social optimum and the golden-rule level of capital?
   d) Determine the slope of the saddle path using Taylor approximation and the rate of adjustment.

4. Suppose $Y_t = F(K_t, A_tL_t)$, with having constant returns to scale and the intensive form of the production function satisfying the Inada conditions. Again, $A_{t+1} = (1+g) A_t$, $L_{t+1} = (1+n) L_t$ and $k_{t+1} = L_t (w_tA_t - C_{1t})$. Assume time is discrete rather than continuous and each individual lives for only two periods. [4+3+2+5]

   a) Solve the individual's maximization problem where θ determines how much individuals varies in response to differences between r (real rate of return) and ρ (discount rate).
   b) Find the ~~speed~~ of convergence of k to k*. How does it differ from Solow model?
   c) Describe how each of the following affects k_{t+1} as a function of k_t:
      (i) A rise in n.
      (ii) A fall in ρ.
   d) After relaxing the assumptions of logarithmic utility and Cobb-Douglas production, what are the possibilities for the relationship between k_t and k_{t+1}? Is there any possibility of dynamic inefficiency?

5. Consider a simplified version of the models of R&D and growth model developed by P.Romer (1990), Gross and Helpman (1991) and Aghion and Howitt (1992) where θ reflects the effect of the existing stock of knowledge on the success of R&D and B is a shift parameter. Both the goods producing sector and R&D producing sector use the full stock of knowledge A. [2+4+...]

a) Critically explain the role of a_L (the labor force used in the R&D sector) for the dynamics of knowledge accumulation without considering capital into the model. Is there any long-run growth effect for change in a_L?

b) Assume that $\theta + \beta < 1$ and $n > 0$, and that the economy is on its balanced growth path. Describe how each of the following changes affects the $\dot{g}_A = 0$ and $\dot{g}_K = 0$ lines and the position of the economy in (g_A, g_K) space at the moment of the change:

(i) A decrease in n.

(ii) An increase in a_K.

c) There are two cases $\beta+\theta<1$ or $\beta+\theta=1$ and $n=0$; What is the difference on the dynamics of the growth rates of capital and knowledge between these two cases?

d) Briefly explain the nature of Knowledge and the determinants of the allocation of resources to R & D including Learning by Doing.

6. Suppose firm i's output is $Y = [\int_{i=0}^{A} L(i)^{\theta} di]^{1/\phi}$ and $L(i)$ is used to denote both the quantity of labor devoted to producing input i and the quantity of input i that goes into final-goods production. Individual's discount rate is ρ and their instantaneous utility function is logarithmic. The utility of the representative household takes the constant-relative-risk aversion (CRRA) form is $\frac{\dot{C}(t)}{C(t)} = \frac{r(t)-\rho}{\theta}$ and θ is the coefficient of the relative risk aversion. Knowledge creation, $\frac{\dot{A}(t)}{A(t)} = BL_A$ .Growth rate of the wage equals the growth rate of output. Population growth is zero. [3+5+3+3]

a) Briefly discuss the assumptions to construct the model.

b) How long-run growth will be determined by using the present value of profits from the discovery of a new idea at time t? What will be the equilibrium value of L_A?

c) What will be the socially optimal level of L_A? Explain how it is different from the equilibrium level of L_A.

d) Briefly explain the implications of this model and extensions of this model.

7. Consider the infinitely lived representative household that derives utility from both consumption (c_t) and leisure ($1-l_t$). The production function of the representative firm is given by [3+2+4+5]

$$f(k_t, l_t, z_t) = z_t k_t^{\alpha} l_t^{1-\alpha}$$

$$z_{t+1} = \varphi z_t + \varepsilon_{t+1}$$

where z_t is the productivity parameter and ε is a white noise error term.

a) Construct the household's problem and using log utility derive the short-run labor supply curve. Explain the relationship among leisure, real wage, and interest rate.

b) Prove that long run labor supply is independent of wage when $\varphi = \frac{1}{3}$

c) Find the elasticity of labor supply and interpret it.

d) Discuss Hansen's indivisible labor supply solution as an attempt to overcome the limitations of previous model and calibration mechanism.

8. The representative producer of good i has production function $Q_i=L_i$; utility depends on consumption and Labor effort where P is known as perfect information. z_i (good specific shock) and m (monetary shock) are normally distributed. Conditional expectation is a linear equation. On the other hand, [4+3+4+3]

For representative consumer $\dot{C}_t = \sigma (r_t - p) C_t$ and Wicksellian interest rate $r^n \equiv p + \sigma^{-1} g$. According to Yun (1996) Optimal price v_t is

$v_t = p_t + \alpha \int_t^{\infty} [(v_s - p_s) + \eta x_s] e^{-(\alpha+p)(s-t)} ds$

and $\kappa \equiv \alpha^2 \eta$; where η is sensitivity parameter.

a) Explain the role of rational expectation in determining the effectiveness of macro policy when there is imperfect information by using Lucas' supply curve.

b) Define New Keynesian DSGE model. Determine the new Keynesian IS equation for continuous time.

c) Determine new Keynesian PC curve for continuous time using Leibniz's rule. Explain "divine coincidence" according to Blanchard and Gali.

d) Describe Taylor rule in the canonical New Keynesian model where x (output) and π (inflation) both are jump variables.

Department of Economics
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M.Sc. Final Examination 2023
Course No: 503 (Development Economics)
Full Marks: 70 Time: Four Hours

*Answer any **five** of the following questions.*
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1. a) What is endogeneity in regression analysis, and how does it arise? 2

b) Suppose you are estimating the effect of health on labor market outcomes with the research question: "Does better health improve an individual's labor market productivity (e.g., employment status or income)?"

(i) Explain how health could be an endogenous variable in this context. 2

(ii) Explain how the Instrumental Variables (IV) method can be used to address and resolve the endogeneity problem in estimating the effect of health on labor market outcomes. 2

c) Suppose you are examining the relationship between health and income across different points of the income distribution.

(i) Why might ordinary least squares (OLS) regression be inadequate in capturing this relationship? 2

(ii) What is Quantile Regression? How can quantile regression provide deeper insights into the impact of health on income across different income levels? 3

d) What is Canonical Correlation Analysis (CCA), and how is it used to examine the relationship between two sets of variables? 3

2. a) Compare the logit and probit models in binary outcome analysis. What are their key similarities and differences? 4

b) In what situations is the multinomial logit model more appropriate than the binary logit model? Explain. 2

c) Answer the following questions:

(i) A logit regression estimates the impact of education and experience on the probability of being employed. The coefficient of education is 0.5. Interpret this coefficient. What does it imply about the relationship between education and employment probability? 2

(ii) In the same logit regression model, the odds ratio for experience is estimated to be 1.7. Interpret this odds ratio. 1

(iii) In a probit model, the marginal effect of income on the probability of owning a house is 0.12. What does this marginal effect mean? 2

(iv) In a multinomial logit model analyzing employment status, the outcome variable has three categories: Employed, Self-employed, and Unemployed (used as the base category). The estimated coefficient for "Male" in the "Self-employed" category is 1.2. Interpret this coefficient in the context of the model. Also, calculate and interpret the corresponding odds ratio. 3

3. a) What do you mean by autoregressive and distributed lag model? Explain with examples. 2

b) What is the ad hoc approach to estimate distributed lag models? Does this approach have any limitations? 4

c) Describe the kyock approach which is applied to distributed lag models? Is this an appropriate approach to convert a distributed lag model to an Autoregressive model? 8

4. a) Discuss the potential limitations of the Adaptive Expectations Model and the Partial Adjustment Model in the context of estimating distributed lag models. 4

b) What role does the Instrumental Variables (IV) method play in overcoming the limitations of estimating Koyck and Adaptive Expectations models? 2

c) Explain the Almon Polynomial Approach to estimating distributed lag models. How does it differ from the Koyck transformation and other distributed lag models? Discuss the key assumptions underlying the Almon approach, along with its advantages and limitations in practical econometric analysis. 8

5. a) Define stationarity in the context of time series analysis. Why is it important in econometric modeling? 3

b) What is unit root problem? How does the Dickey-Fuller (DF) test help detect the presence of a unit root in a time series? Explain the testing procedure. 5

c) Critically explain the following unit root tests: 6

(i) Augmented Dickey-Fuller (ADF) unit root test

(ii) Phillips-Perron (PP) unit root test, and

(iii) Kwiatkowski–Phillips–Schmidt–Shin (KPSS) unit root test

6. a) Why do structural breaks matter in unit root testing? Briefly explain their impact on standard unit root tests. 3

b) What is the Zivot-Andrews test for unit root? Describe its procedure and how it accounts for structural breaks. 4

c) What is cointegration in time series analysis? How can we test whether two or more variables are cointegrated? Explain. 4

d) Why might a researcher prefer the ARDL bounds testing approach over the Johansen cointegration test? Discuss the key differences that guide this choice. 3

7. a) Consider the ARIMA (2,1,1) model. Write down its equation and explain its components in plain terms. 2

b) How does the Box-Jenkins methodology guide the estimation and evaluation of ARIMA models? Explain briefly. 4

c) Suppose you have two variables: inflation and unemployment. How could a VAR model help you understand their relationship over time? Explain. 4

d) Define Granger causality and explain how it can be tested using a VAR model. 4

8. a) What are the advantages of panel data over cross-section and time series data? 3

b) What is the Fixed Effect Least-Squares Dummy Variable (LSDV) model in panel data analysis? Explain how it controls for unobserved heterogeneity and discuss its limitations. 5

c) What is the Panel Random Effects Model? How does it differ from the Fixed Effects Model in terms of assumptions and estimation? When is it appropriate to use the Random Effects approach? 6

Department of Economics
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M.Sc. Final Examination 2023
Course No: 507 (Health Economics)
Full Marks: 70 Time: Four Hours

*Answer any **five** of the following questions.*
All parts of a question must be answered consecutively

1. What is the difference between affordable care act (ACA) and federal level dependent coverage? According to Da Jung Jun (2023) study, briefly explain the effects of dependent health insurance coverage mandates on father's job mobility and compensation. 14

2. "Long Covid significantly impacts productivity losses and provision of informal care, exacerbated by high national prevalence of long Covid". Justify this research result in the light of Kwon et al., (2023) study. 14

3. Briefly discuss the Louisiana family planning program (LAFPP). According to Gamino (2023) study, briefly explain the effect of a free family planning program on fertility. 14

4. Is supported employment effective for disability insurance recipients with mental health conditions? Discuss your answer in the light of Fontenay and Tojerow (2025) studies. 14

5. Briefly discuss the effects of the minimum wage on child health, in the light of Wehby et al., (2022) research. 14

6. a) "In a perfect health insurance market, a risk-averse individual will always be willing to pay full and fair premium for health insurance." Discuss. 6

   b) Show that in case of adverse selection:

   i. Pooling equilibrium is not sustainable 3

   ii. Separating equilibrium is possible only if the proportion of high-risk individual's is much lower than the low-risk individual in the pool. 5

7. How can you define triage of the emergency departments for medical care? Briefly discuss the importance of the emergency departments and identify the challenges in Eds. 14

8. Based on the Holford and Rabe (2024) studies, briefly discuss the impact on children's body weight of switching from means-tested to universal provision of nutritious free school meals. 14

Department of Economics
Jahangirnagar University
M.Sc. Final Examination 2023
Course No: 510 (Labor Economics)
Full Marks: 70 Time: Four Hours

*Answer any **five** of the following questions.*

<u>All parts of a question must be answered consecutively</u>

1. a) Sketch the circular flow diagram to demonstrate the linkage between the labor market and the mainstream economy. [3]

   b) Briefly explain the distinctive features of the labor market which segregate this market from other types of market. [6]

   c) Why does the employer set work schedules to signify an institutional restraint on individual choice? – Explain using a diagram. [5]

2. Suppose an employer pays a specific wage rate per hour when the total hours worked in a week are equal to or less than 40 hours. For every extra hour over 40 hours per week, the wage rate is 2 (two) times the above wage rate. Now use the labor/leisure model to answer the following questions:

   a) First, draw a budget constraint for someone earning $10 per hour who receives $200 a week in non-labor income. What happens to the budget constraint with overtime laws? [3]

   b) Given your answer to (a), draw an indifference curve so that, given the original budget constraint, the person is working 40 hours per week. What now happens to hours of work, given the new budget constraint? Again, sketch an indifference curve to show this. Did work hours increase or decrease due to the overtime law? Explain why. [3]

   c) In words, define the income and substitution effects of a wage change. Use diagram(s) to identify the income and substitution effects. Which one is more robust? [4]

   d) If the wage increases for all hours worked rather than just for those after 40 hours per week, would the change in hours of work be any different? Again, show this in the graph and explain why or why not. [4]

3. a) Data from a recent study indicates that 32% of salaried workers work 45 hours or more (instead of 40 hours) per week. What can account for this apparent discrepancy between theory and fact? [7]

   b) There is a debate about whether the size of the labor force will expand or contract during periods of recession and depression. - explain the debate with the help of added worker effect and discourage worker effect. [7]

4. a) "Time and monetary costs associated with working affect the shape of budget constraint thus influence the number of hours a person chooses to work" – Explain this statement using appropriate diagrams. [8]

   b) Why do policymakers initiate wage subsidy programs? Illustrate the effect of a wage subsidy program on the labor market. [6]

5. a) How is the equilibrium level of employment determined in the long run? Why is the long-run labor demand curve more elastic than the short-run labor demand curve? [7]

b) Despite the revolution in technology in the twentieth century, however, the number of jobs in the economy has not decreased rather has increased by many millions. How can this be accounted for? Explain the impact of technological change on employment with the long-run theory of labor demand. [7]

6. a) Describe the wage determination model of a monopsony market. Illustrate how the monopsonist exploits labor. [6]

b) What are the positive and negative aspects of minimum wage law in a competitive market? Is it possible to obtain competitive outcomes if the government imposes a minimum wage law in a monopsony market? [8]

7. a) Illustrate the investment decision rules for taking higher education using age/earnings profile. [7]

b) Differentiate between general on-the-job training and specific on-the-job training. Graphically analyze the benefits and costs of these two training. [7]

8. a) How can years of education be used as a good screening device in the employee selection process? Discuss using diagram(s) the signaling failure and the optimal signal when education is used as a screening device. [8]

b) Briefly explain different factors that influence the achievement of the external alignment of pay level. [3]

c) It has been suggested that one way to raise the quality of teaching at colleges and universities is to link professors' pay to their classroom performance, say by making pay raises a positive function of the score they receive on end-of-semester student course evaluations. Evaluate the pros and cons of this pay form. [3]